

1. What is CSS?

- CSS stands for Cascading Style Sheets.
- It is used to style and design HTML web pages.
- It controls the look and feel of websites (colors, fonts, spacing, layouts etc.).

Example :

```
h1 {
  color: blue;
  font-size: 28px;
}
```

This makes all
→ **<h1>** headings
blue and size
28px.

Benefits of CSS :

- ✓ Makes websites attractive and professional.
- ✓ Saves time - write once, use multiple times.
- ✓ Easy to maintain and update.
- ✓ Improves user experience.

2. Types of CSS

★ Inline CSS

- Used inside HTML tags using **style** attribute.
- Affects only that particular element.

Example :

```
<h1 style="color:red;
font-size:20px;">
Hello World
</h1>
```

This will make this heading **red** and size 20px.

★ Internal CSS

- Used inside **<style>** tag in the **<head>** section.
- Affects all elements in the page.

Example :

```
<head>
<style>
h1 {
  color: blue;
  font-size: 24px;
}
</style>
</head>
```

This style will apply to all **<h1>** in this page.

★ External CSS

- Used in a separate .css file.
- Linked to HTML using **<link>** tag
- Best for large websites.

Example :

```
<head>
<link rel="stylesheet"
href="style.css">
</head>
```

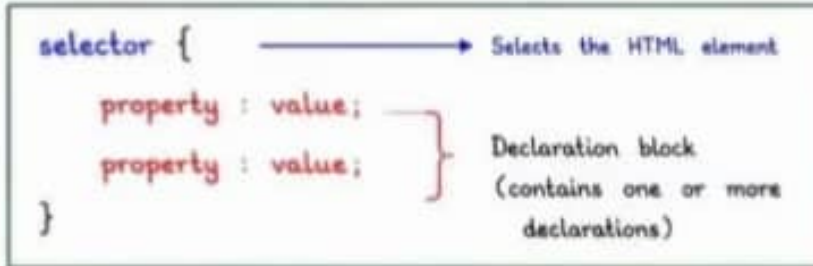
All styles written in **style.css** file will be applied to this page.

Comparison Table

Type	Where it is Written	Scope	Best For
Inline CSS	Inside HTML tag (style attribute)	Single element	Small changes
Internal CSS	Inside <style> tag in <head>	Whole page	Medium websites
External CSS	Separate .css file	Multiple pages	Large websites

3. CSS Syntax

Each CSS rule contains a **selector** and a **declaration block**.



Example :

```
p {
  color: green;
  font-size: 18px;
}
```

→ All `<p>` elements will be green.

→ Font size of all `<p>` will be 18px.

Syntax Components

- **Selector** : Targets the HTML element.
- **Property** : The style you want to change.
- **Value** : The value of the property.
- **Declaration** : A property and its value.

4. CSS Selectors

Selectors are used to select the HTML elements you want to style.

Example in HTML

```
<h1 class="title">Hello</h1>
<p id="intro">Welcome to CSS Notes</p>
<p>This is a paragraph.</p>
```

Selector	Syntax	Description	Example	Selects
1. Element Selector	<code>element { }</code>	Selects all elements of that type.	<code>p { color: blue; }</code>	All <code><p></code> elements
2. Class Selector	<code>.class { }</code>	Selects all elements with a specific class.	<code>.title { color: red; }</code>	All elements with <code>class="title"</code>
3. ID Selector	<code>#id { }</code>	Selects the element with a specific id.	<code>#intro { font-size: 18px; }</code>	Element with <code>id="intro"</code>
4. Universal Selector	<code>* { }</code>	Selects all elements on the page.	<code>* { margin: 0; }</code>	Every element on the page
5. Group Selector	<code>h1, p { }</code>	Selects multiple elements together.	<code>h1, p { color: purple; }</code>	All <code><h1></code> and all <code><p></code> elements

★ Note :

- ✓ ID should be unique in a page.
- ✓ Class can be used multiple times.
- ✓ You can combine selectors for advanced styling.

Example (Grouping) :

```
h1, h2, h3 {
  color: darkblue;
}
```

→ Applies same style to `h1`, `h2` and `h3`.

5. CSS Colors

Colors are used to set the **text color**, **background color**, **border color**, etc.

★ Ways to Specify Colors :

- 1. Color Name → red, blue, green, etc.
- 2. HEX Value → #RRGGBB
- 3. RGB Value → rgb(red, green, blue)
- 4. HSL Value → hsl(hue, saturation%, lightness%)

Examples :

- **color: red;** /* Color Name */
- **color: #1E90FF;** /* HEX Value */
- **color: rgb(0, 128, 0);** /* RGB */
- **color: hsl(120, 60%, 50%);** /* HSL */

6. CSS Backgrounds

Background properties are used to set the **background color**, **image**, **position**, **repeat**, etc.

Example :

```
body {
  background-color: #f0f8ff;
  background-image: url('bg.jpg');
  background-repeat: no-repeat;
  background-position: center;
  background-size: cover;
}
```

Example Output :



← Background image is set, centered and covers the whole area.

Common Background Properties :

Property	Description
background-color	Sets the background color
background-image	Sets the background image
background-repeat	Repeat the background image
background-position	Sets the starting position of the background image
background-size	Sets the size of the background image
background-attachment	Sets whether background scrolls with the page or not

7. CSS Borders

Borders are used to add a **border** around elements.

Example :

```
div {
  border: 2px solid blue;
  border-radius: 10px;
}
```

Example Output :

This is a box with border.

Border Styles :



★ Note :

- ✓ Color names are not case-sensitive (Red = red = RED).
- ✓ HEX values must start with #.
- ✓ Use **background-size: cover;** for full screen background images.
- ✓ **border-radius: 50%** makes the element circular (if width = height).

Common Border Properties :

Property	Description
border	Shorthand property for border-width, style and color
border-width	Sets the thickness of the border
border-style	Sets the style of the border (solid, dashed, dotted, etc.)
border-color	Sets the color of the border
border-radius	Rounds the corners of the border

Example (Circle) :

```
div {
  width: 100px;
  height: 100px;
  background: orange;
  border-radius: 50%;
}
```

Output :



8. CSS Text & Fonts

Text properties are used to style the text content.

Example :

```
p {
  color: darkblue;
  font-family: 'Poppins', sans-serif;
  font-size: 18px;
  text-align: center;
  text-decoration: underline;
  text-transform: capitalize;
  line-height: 1.6;
}
```

★ Note :

Fonts can be from system fonts or web fonts (Google Fonts).

Common Text Properties :

Property	Description	Example
color	Sets the text color	color: red;
font-family	Sets the font of text	font-family: Arial, sans-serif
font-size	Sets the size of text	font-size: 20px;
font-weight	Sets thickness of text	font-weight: bold;
font-style	Sets style of text	font-style: italic;
text-align	Aligns text (left, center, right, justify)	text-align: center;
text-decoration	Adds decoration (none, underline, overline, line-through)	text-decoration: underline;
text-transform	Changes text case (uppercase, lowercase, capitalize)	text-transform: uppercase;
line-height	Sets space between lines	line-height: 1.8;
letter-spacing	Sets space between letters	letter-spacing: 2px;
word-spacing	Sets space between words	word-spacing: 5px;

9. CSS Lists & Tables

★ A. List Styles

Used to remove or change the default style of lists.

Example :

```
ul {
  list-style-type: square;
  list-style-position: inside;
}
```

Common list-style-type values :

Value	Example
disc	● Item
circle	○ Item
square	■ Item
none	Item (no bullet)
decimal	1. Item
upper-roman	I. Item
lower-alpha	a. Item

★ B. Table Styles

Used to style HTML tables.

Example :

```
table {
  width: 100%;
  border-collapse: collapse;
}
th, td {
  border: 1px solid #4CAF50;
  padding: 10px;
  text-align: center;
}
```

Example Output :

Name	Course	City
Sagar	BCA	Jaipur
Rahul	MCA	Delhi
Neha	B.Sc IT	Pune

★ Note :

- Use `border-collapse: collapse;` to remove double borders in tables.
- `padding` adds space inside table cells.



Quick Tip : Good text styling improves readability and makes your website look professional.

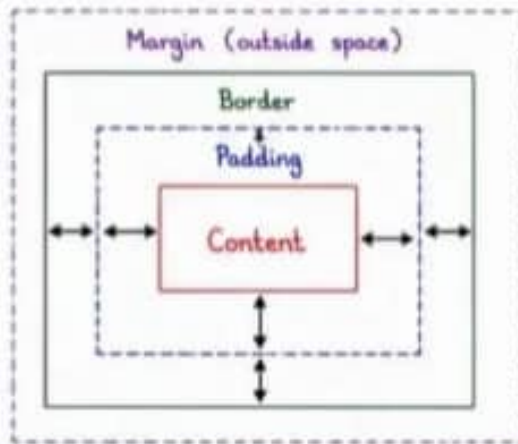


10. CSS Box Model

Every HTML element is a rectangular box.

It consists of :

1. Content
2. Padding
3. Border
4. Margin



Example :

```
div {
  width: 300px;
  padding: 20px;
  border: 2px solid blue;
  margin: 15px;
}
```

★ Note :

Margin is outside the border.
Padding is inside the border around the content.

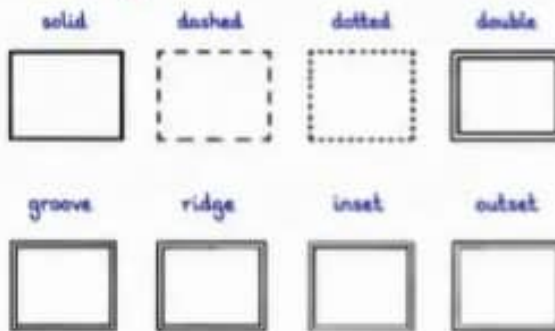
11. CSS Borders

Borders are used to define the boundary of an element.

Example :

```
p {
  border-width: 2px;
  border-style: solid;
  border-color: red;
}
```

Border Styles :



Border Shorthand :

border: 2px dashed green;
 ↑ ↑ ↑
 width style color

★ Note :

If border-width is not given, it is 3px by default.

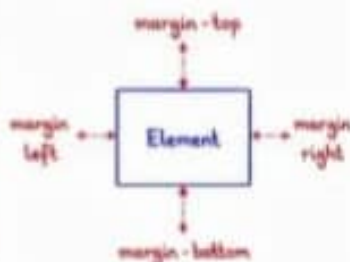
12. CSS Margins & Padding

Margins (outside space)

Margins create space outside the border.

Example :

```
div {
  margin-top: 20px;
  margin-right: 15px;
  margin-bottom: 20px;
  margin-left: 15px;
}
```

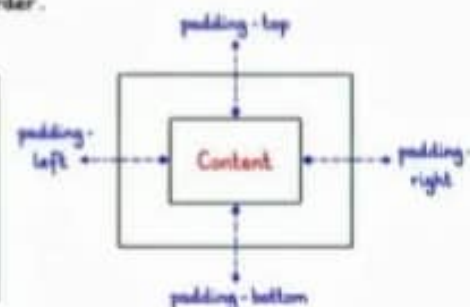


Padding (inside space)

Padding creates space inside the border between content and border.

Example :

```
div {
  padding-top: 15px;
  padding-right: 10px;
  padding-bottom: 15px;
  padding-left: 10px;
}
```



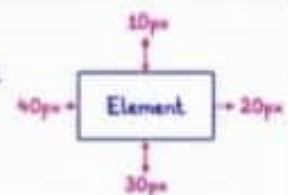
★ Note :

- ✓ You can also use shorthand.
- ✓ margin: 20px; (all four sides)
- ✓ padding: 10px 20px; (top & bottom 10px, left & right 20px)

Shorthand Order (4 values) :

margin / padding: top right bottom left

Example: margin: 10px 20px 30px 40px;



13. CSS Display Property

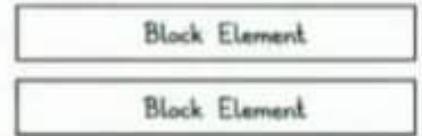
The **display** property specifies how an element is displayed on a webpage.

Common Values :

Value	Description	Example
block	Takes full width. Starts on a new line.	<code>div { display: block; }</code>
inline	Takes only required width. Does not start on new line.	<code>span { display: inline; }</code>
inline-block	Behaves like inline but we can set width & height.	<code>div { display: inline-block; }</code>
none	Hides the element. Not visible on the page.	<code>p { display: none; }</code>
flex	Used for flexible box layout.	<code>div { display: flex; }</code>
grid	Used for grid layout.	<code>div { display: grid; }</code>

Visual Representation :

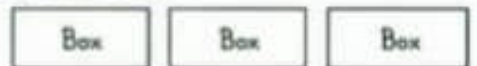
1. block



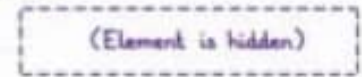
2. inline



3. inline-block



4. none



★ Note :

By default,

- `div`, `h1` to `h6`, `p`, `form`, `header`, `footer` → `display: block`
- `span`, `a`, `img`, `label` → `display: inline`

Example :

```

box {
  display: inline-block;
  width: 100px;
  height: 50px;
  background-color: lightblue;
  margin: 5px;
  text-align: center;
  line-height: 50px;
}

```

Output :



14. CSS Position Property

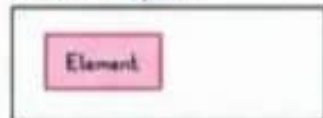
The **position** property specifies the type of positioning method used for an element.

Common Values :

Value	Description	Example
static	Default position. Follows normal flow.	<code>div { position: static; }</code>
relative	Positioned relative to its normal position.	<code>div { position: relative; top: 10px; left: 20px; }</code>
absolute	Positioned relative to nearest positioned ancestor. Removed from normal flow.	<code>div { position: absolute; top: 0; right: 10px; }</code>
fixed	Fixed position relative to browser window. Does not move on scroll.	<code>div { position: fixed; bottom: 20px; right: 20px; }</code>
sticky	Acts like relative until a certain scroll position, then sticks like fixed.	<code>div { position: sticky; top: 0; }</code>

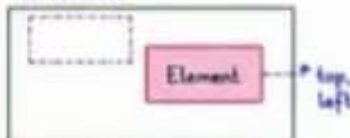
Visual Representation :

1. static (default)



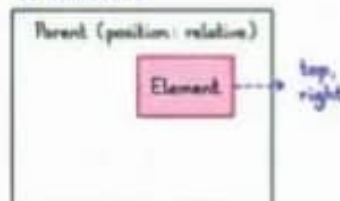
Normal position in document flow.

2. relative



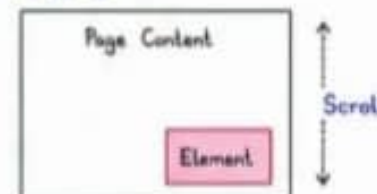
Moves relative to its normal position.

3. absolute



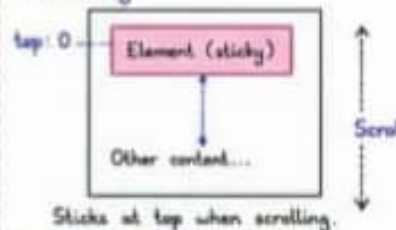
Positioned relative to parent.
Removed from normal flow.

4. fixed



Stays fixed in browser window.
Does not move on scroll.

5. sticky



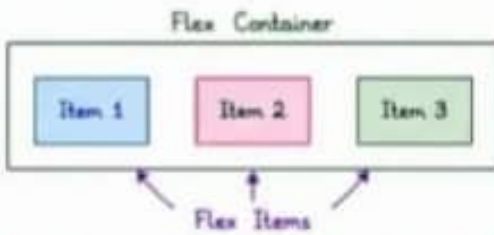
Sticks at top when scrolling.

★ Note :

For absolute, fixed & sticky,
we can use: `top`, `right`,
`bottom`, `left`.

15. CSS Flexbox (Basics)

Flexbox is a one-dimensional layout method for arranging items in rows or columns.



To use flexbox :

```
.container {
  display: flex;
}
```

All direct children become flex items.

Anatomy of Flexbox



Flex Container Properties (Most Important)

Property	Description	Values	Example
flex-direction	Defines the direction of items.	row row-reverse column column-reverse	flex-direction: row;
justify-content	Aligns items along main axis.	flex-start flex-end center space-between space-around space-evenly	justify-content: center;
align-items	Aligns items along cross axis.	stretch flex-start flex-end center baseline	align-items: center;
flex-wrap	Allows items to wrap to next line.	nowrap wrap wrap-reverse	flex-wrap: wrap;
align-content	Aligns multiple lines (when wrap).	stretch flex-start flex-end center space-between space-around space-evenly	align-content: space-between;

Visual Examples :

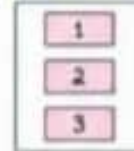
1. flex-direction: row; (default)



2. flex-direction: row-reverse;



3. flex-direction: column;



4. justify-content: space-between;



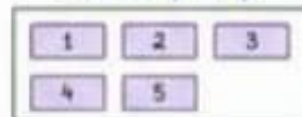
5. justify-content: center;



6. align-items: center;



7. flex-wrap: wrap;



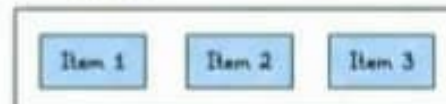
Flex Item Properties

Property	Description	Example
flex-grow	How much item will grow.	flex-grow: 1;
flex-shrink	How much item will shrink.	flex-shrink: 1;
flex-basis	Initial size of item.	flex-basis: 200px;
flex	Shortcut for grow, shrink, basis.	flex: 1 1 200px;
align-self	Align single item (override align-items).	align-self: flex-end;

Example Code :

```
container {
  display: flex;
  justify-content: space-around;
  align-items: center;
  height: 150px;
  border: 2px solid #4CAF50;
}
.item {
  background: skyblue;
  padding: 15px 25px;
}
```

Output :



★ Note :

Flexbox is perfect for building menus, cards, navigation bars, centering elements and many modern layouts.



That's it for CSS Basics! 🎁

Practice more & build amazing websites. Happy Coding! 🚀

@ Mastercode snnr

Remember :

"Code + Practice"
= Skill + Confidence" ❤️